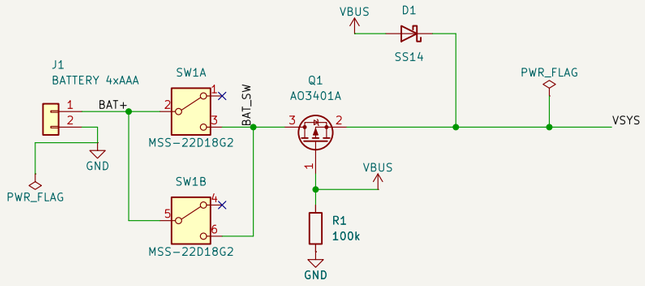
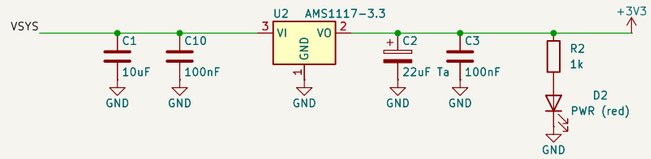
1. POWER INPUT & PROTECTION

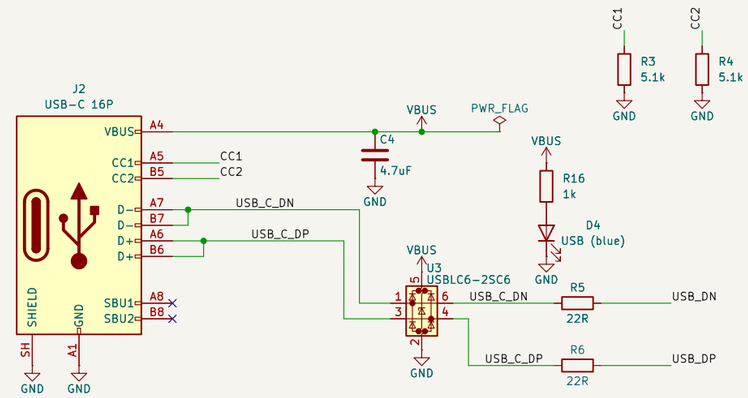


2. 3.3V LDO REGULATOR

AMS1117

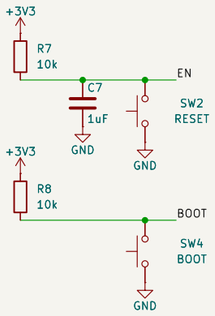


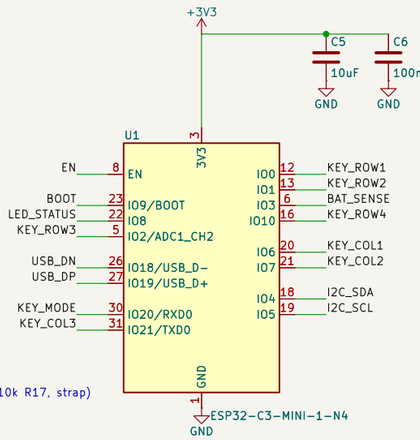
3. USB-C: PROGRAMMING / DEBUG / 5V



4. MCU: ESP32-C3-MINI-1-N4

RESET / BOOT

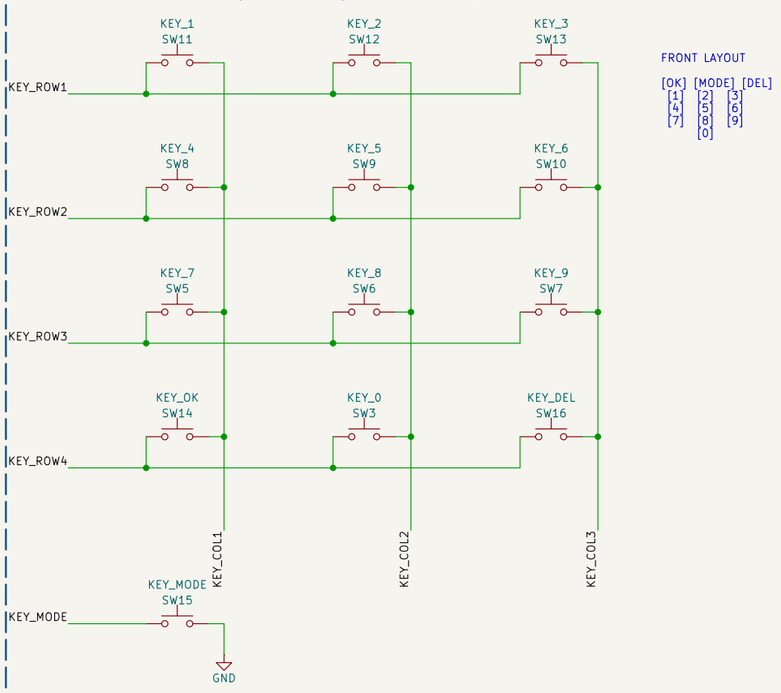




GPIO MAP
IO0/IO1/IO2/IO10 KEY_ROW1..4 (input pull-up; IO2 +10k R17, strap)
IO6/IO7/IO21 KEY_COL1..3 (scan, drive LOW)
IO20 KEY_MODE (direct, pull-up)
IO4/IO5 I2C SDA/SCL (OLED)
IO8 LED_STATUS (active LOW, strap=1)
IO9 BOOT button (strap)
IO3 BAT_SENSE (ADC1_CH3)
IO18/IO19 USB D-/D+

5. KEYPAD: 4x3 MATRIX + MODE

13x tact switch 6x6mm SMD H7mm (PTS645 footprint) | rows = inputs w/ pull-up, cols = scan outputs

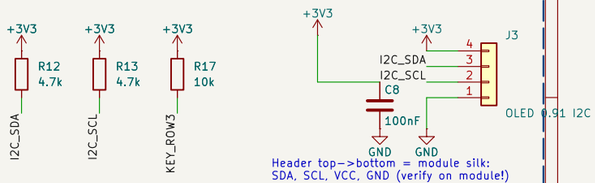


FRONT LAYOUT

[OK] [MODE] [DEL]
[1] [2] [3]
[4] [5] [6]
[7] [8] [9]
[0]

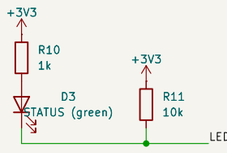
6. OLED / STATUS LED / BATTERY SENSE

OLED 0.91" SSD1306 128x32 off-board via 4-pin header J3
J3 order: 1 GND / 2 SCL / 3 SDA / 4 3V3 (jumper wires)

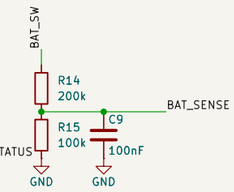


Header top->bottom = module silk:
SDA, SCL, VCC, GND (verify on module!)

STATUS LED (IO8)



BATTERY SENSE (IO2 ADC)



MECHANICAL



DESIGN NOTES
1. Board 70 x 90 mm

Ref designs: Espressif ESP32-C3-MINI-1 , AMS1117
PCB 70x90 mm, single-sided
Blocks: 1 Power input | 2 LDO 3.3V | 3 USB-C | 4 MCU | 5 Keypad | 6 OLED/LED/Battery sense
HK1 NAM3 — Mon IoT

Sheet: /
File: IOT.kicad_sch

Title: IoT Classroom Clicker — ESP32-C3 Student Keypad

Size: A3	Date: 2026-09-29	**Rev: 1.0**
KiCad E.D.A. 10.0.5-rc1		Id: 1/1